



Algebra and Functions · Mark Scheme

Chapter 2: Algebra and Functions. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Indices and surds

- 1 $27^{-\frac{2}{3}} = \frac{1}{(\sqrt[3]{27})^2}$ M1 = $\frac{1}{9}$ A1.
- 2 $\sqrt{75} = 5\sqrt{3}$, $\sqrt{12} = 2\sqrt{3}$, $\sqrt{48} = 4\sqrt{3}$ M1 A1 (at least two correct); total $7\sqrt{3}$ A1.
- 3 Multiply by $\frac{\sqrt{5} + \sqrt{2}}{\sqrt{5} + \sqrt{2}}$ M1; denominator $5 - 2 = 3$ A1; $\frac{6(\sqrt{5} + \sqrt{2})}{3} = 2\sqrt{5} + 2\sqrt{2}$ A1.
- 4 Split: $\frac{x^2}{2x^{1/2}} + \frac{4}{2x^{1/2}}$ M1; $= \frac{1}{2}x^{\frac{3}{2}} + 2x^{-\frac{1}{2}}$ A1 A1 (one per term).
- 5(a) $4^x = (2^2)^x = y^2$ B1; $2^{x+3} = 2^3 \cdot 2^x = 8y$ B1.
- 5(b) $y^2 + 8y - 20 = 0$ M1; $(y + 10)(y - 2) = 0 \Rightarrow y = 2$ (reject $y = -10$ since $2^x > 0$) A1; $2^x = 2 \Rightarrow x = 1$ A1.

Section B – Quadratics

- 6 $(2x - 1)(x - 3) = 0$ M1; $x = \frac{1}{2}$ or $x = 3$ A1.
- 7 $(x + 4)^2 - 11$ M1 A1; minimum point $(-4, -11)$ B1 ft.
- 8 $b^2 - 4ac = (-4)^2 - 4(3)(5)$ M1 = -44 A1; negative, so no real roots A1.
- 9(a) $-3(x^2 - 4x) + 5$ M1; $-3[(x - 2)^2 - 4] + 5$ M1; $17 - 3(x - 2)^2$, so $a = 17$, $b = 3$, $c = -2$ A1.
- 9(b) Maximum value 17 when $x = 2$ B1 ft.
- 10 $b^2 - 4ac < 0$: $(4k)^2 - 4(k)(3) < 0$ M1; $16k^2 - 12k < 0$ A1; critical values $k = 0$, $\frac{3}{4}$ M1; $0 < k < \frac{3}{4}$ A1. Check $k = 0$: equation becomes $3 = 0$, which has no roots, so include it: $0 \leq k < \frac{3}{4}$ B1.
Teaching point: this final mark separates grade A from A*; the discriminant only applies when the equation is a genuine quadratic ($k \neq 0$).

Section C – Simultaneous equations

- 11 Eliminate y : $6x + 4y = 14$, add to $5x - 4y = 19$ M1; $11x = 33$, $x = 3$ A1; $y = -1$ A1.
- 12 $x + 1 = x^2 - 5 \Rightarrow x^2 - x - 6 = 0$ M1 A1; $x = 3$ or $x = -2$ A1; $(3, 4)$ and $(-2, -1)$ A1.
- 13 $x = y + 4$ M1; $(y + 4)^2 + y(y + 4) = 6$ M1; $2y^2 + 12y + 10 = 0$ A1; $y = -1$ or $y = -5$ A1; $x = 3$, $y = -1$ and $x = -1$, $y = -5$ A1.
- 14(a) $x^2 + 3 = 2x + k \Rightarrow x^2 - 2x + (3 - k) = 0$ M1; tangent so $b^2 - 4ac = 0$: $4 - 4(3 - k) = 0$ M1; $k = 2$ A1.
- 14(b) $x^2 - 2x + 1 = 0 \Rightarrow x = 1$ M1; point $(1, 4)$ A1.
- 15 $x = 2 - y$ M1; $4y^2 - (2 - y)^2 = 11$ M1; $4y^2 - 4 + 4y - y^2 = 11$ M1 (expand); $3y^2 + 4y - 15 = 0$ A1; $(3y - 5)(y + 3) = 0$ M1; $y = \frac{5}{3}$ or $y = -3$ A1; $x = \frac{1}{3}$, $y = \frac{5}{3}$ and $x = 5$, $y = -3$ A1.

Section D – Inequalities

- 16 $2x < 12$ M1; $x < 6$ A1.
- 17 $6 - 3x \geq 5x + 22$ M1; $-16 \geq 8x$ M1; $x \leq -2$ A1.
- 18 $(2x - 1)(x + 3) > 0$ M1; critical values $\frac{1}{2}$, -3 A1; outside region chosen M1; $x < -3$ or $x > \frac{1}{2}$ A1.
- 19(a) $x > 3$ B1. 19(b) $(x - 2)(x - 5) < 0$ M1; critical values 2, 5 A1; $2 < x < 5$ A1. 19(c) $\{x : 3 < x < 5\}$ B1 ft.



20(a) $kx - 2 = x^2 + 2x + 2 \Rightarrow x^2 + (2 - k)x + 4 = 0$ M1 A1; no intersection so $b^2 - 4ac < 0$: $(2 - k)^2 - 16 < 0 \Rightarrow (2 - k)^2 < 16$ (given answer, must be convincing) A1.

20(b) $-4 < 2 - k < 4$ or $k^2 - 4k - 12 < 0$ M1; critical values $-2, 6$ A1; $-2 < k < 6$ A1.

Section E – Factor and remainder theorems

21 $f(2) = 16 - 12 + 2 - 5$ M1 = 1 A1.

22 $f(-3) = -27 + 18 + 15 - 6 = 0$ M1; so $(x + 3)$ is a factor A1 – conclusion needed.

23 $f(-1) = -1 + 7 - 6 = 0$, so $(x + 1)$ is a factor M1 A1; $x^3 - 7x - 6 = (x + 1)(x^2 - x - 6)$ M1; $= (x + 1)(x - 3)(x + 2)$ A1.

24 $f(1) = 0$: $1 + a + b + 6 = 0 \Rightarrow a + b = -7$ M1 A1; $f(-1) = 8$: $-1 + a - b + 6 = 8 \Rightarrow a - b = 3$ M1 A1; $a = -2, b = -5$ A1.

25(a) $f(3) = 54 - 45 - 12 + 3 = 0$ M1; so $(x - 3)$ is a factor A1.

25(b) $f(x) = (x - 3)(2x^2 + x - 1)$ M1 A1; $= (x - 3)(2x - 1)(x + 1)$ M1 A1.

25(c) Roots of $f(x) = 0$ are $3, \frac{1}{2}, -1$; $f(x - 1) = 0$ when $x - 1$ equals these M1; $x = 4, \frac{3}{2}, 0$ A1 ft.

Section F – Partial fractions

26 $5x + 1 \equiv A(x - 1) + B(x + 1)$ M1; $x = 1$: $B = 3$; $x = -1$: $A = 2$ A1 A1. $\frac{2}{x + 1} + \frac{3}{x - 1}$.

27 $x + 10 \equiv A(2x - 1) + B(x + 3)$ M1; $x = -3$: $7 = -7A, A = -1$ M1 A1; $x = \frac{1}{2}$: $10.5 = 3.5B, B = 3$ A1. $-\frac{1}{x + 3} + \frac{3}{2x - 1}$.

28 $2x^2 + 6x - 2 \equiv A(x - 1)(x + 2) + Bx(x + 2) + Cx(x - 1)$ M1; $x = 0$: $A = 1$; $x = 1$: $B = 2$; $x = -2$: $C = -1$ A1 A1 A1. $\frac{1}{x} + \frac{2}{x - 1} - \frac{1}{x + 2}$.

29 $x^2 + 7x + 1 \equiv A(x - 1)(x + 2) + B(x + 2) + C(x - 1)^2$ M1; $x = 1$: $9 = 3B, B = 3$ A1; $x = -2$: $-9 = 9C, C = -1$ A1; compare x^2 : $1 = A + C$ M1; $A = 2$ A1.

30 Divide (or compare x^2 coefficients): $A = 3$ B1; $3x^2 + 2x - 2 \equiv 3(x + 2)(x - 1) + B(x - 1) + C(x + 2)$ M1; $x = 1$: $3 = 3C, C = 1$ A1; $x = -2$: $6 = -3B, B = -2$ M1 A1.

Common error: missing the constant A because the fraction is improper.

Section G – Graph transformations

31 (a) $(2, -2)$ B1; (b) $(-1, -6)$ B1.

32 (a) $(-4, 18)$ B1; (b) $(-8, 6)$ B1; (c) $(-4, -6)$, now a minimum B1.

33 (a) Translation by vector $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$ B1. (b) Stretch parallel to the y -axis, scale factor 8 B1; or stretch parallel to the x -axis, scale factor $\frac{1}{2}$ (since $8x^3 = (2x)^3$) B1.

34(a) $y = (x - 3)^2 - 2$ M1; $y = x^2 - 6x + 7$ A1. **34(b)** $y = 2(x^2 - 6x + 7)$ M1; $y = 2x^2 - 12x + 14$ A1.

35(a) U-shaped curve translated 2 left B1; crosses x -axis at $(-2, 0)$ and $(2, 0)$ B1; minimum $(0, -4)$, which is also the y -intercept B1.

35(b) Same roots $(0, 0)$ and $(4, 0)$ B1; minimum $(2, -8)$ B1.

35(c) $k = 4$ B1. **35(d)** $(x + 2)^2 - 4(x + 2) = x^2 - 4$ B1.

Section H – Functions

36 (a) $g(2) = 5, f(5) = 17$ B1; (b) $f(1) = 1, g(1) = 2$ B1.

37 $y(x - 3) = 2x + 1 \Rightarrow xy - 2x = 3y + 1$ M1; $x = \frac{3y + 1}{y - 2}$ A1; $f^{-1}(x) = \frac{3x + 1}{x - 2}, x \in \mathbb{R}, x \neq 2$ A1.



38 (a) $f(x) \geq 2$ B1. (b) $g(-1) = 5, g(4) = -5$ M1; $-5 \leq g(x) \leq 5$ A1.

39(a) $(x-3)^2 + 2$ M1 A1. **39(b)** $f(x) \geq 2$ B1 ft.

39(c) $y = (x-3)^2 + 2 \Rightarrow x = 3 \pm \sqrt{y-2}$ M1; positive root since $x \geq 3$: $f^{-1}(x) = 3 + \sqrt{x-2}$ A1; domain $x \geq 2$ B1 ft.

40(a) $g(x) > 0$ B1.

40(b) $y = \frac{3}{x-1} \Rightarrow x = 1 + \frac{3}{y}$ M1 A1; $g^{-1}(x) = 1 + \frac{3}{x}, x > 0$ B1 ft.

40(c) $fg(x) = \frac{6}{x-1} + 5$ B1; $\frac{6}{x-1} + 5 = x \Rightarrow x^2 - 6x - 1 = 0$ M1 A1; $x = 3 \pm \sqrt{10}$; since $x > 1$, only $x = 3 + \sqrt{10}$ A1 – rejection must be justified.

Section I – The modulus function

41 $x-4 = 7$ or $x-4 = -7$ M1; $x = 11$ or $x = -3$ A1.

42 $-5 < 2x+3 < 5$ M1; $-8 < 2x < 2$ M1; $-4 < x < 1$ A1.

43 $3x-2 \leq -4$ or $3x-2 \geq 4$ M1; $x \leq -\frac{2}{3}$ A1; $x \geq 2$ A1.

44 $x+5 = 3x-1 \Rightarrow x = 3$ M1 A1; $x+5 = -(3x-1) \Rightarrow x = -1$ M1 A1; both check correctly.

45(a) V-shape with vertex (3, 0) and y-intercept (0, 6) B1; straight line through (0, 1) and (-1, 0) B1.

45(b) $2x-6 = x+1 \Rightarrow x = 7$ M1 A1; $-(2x-6) = x+1 \Rightarrow x = \frac{5}{3}$ M1 A1. Both valid since $x+1 > 0$.

45(c) Graph of modulus above the line: outside the roots M1; $\{x : x < \frac{5}{3}\} \cup \{x : x > 7\}$ A1 ft.

Roar-level challenge

46 $x^2 - 5 = 4$ or $x^2 - 5 = -4$ M1; $x^2 = 9 \Rightarrow x = \pm 3$ A1; $x^2 = 1 \Rightarrow x = \pm 1$ A1; all four solutions check in the original equation A1.